UMI AI Chatbot — Comprehensive Technical Documentation

Enterprise System Architecture & Reference Guide | Version 2.0.0 (August 2026)

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1. System Overview & Core Capabilities

UMI AI Chatbot is an enterprise-grade AI assistant web application built with Django 6.0 backend, MySQL (MariaDB) persistent storage, Vanilla Modern Glassmorphic UI, LlamaIndex RAG engine, and dual LLM integrations (Google Gemini & local Ollama).

Key Features & Systems:

- **Dual AI Providers:** Local LLMs via Ollama (llama3.2, qwen2.5) and cloud LLMs via Google Gemini API.
- **Enterprise MySQL Database:** 14 relational tables managing users, settings, sessions, messages, documents, logs, and security tokens.
- **Enterprise Auth Subsystem:** Registration, Login, Logout, Remember Me (30-day session), Password Recovery & Reset via SMTP.
- **LlamaIndex RAG Engine:** Page-aware document extraction and chunking using SentenceSplitter across 25+ document/code file extensions.
- **Profile & Avatar Processing:** Pillow Image Optimization (LANCZOS 300x300 JPG filter), thumbnail previews, and dynamic initial avatar rendering.
- **Atomic Account Cleanup:** Password-verified transactional cascade account deletion removing all files, records, and chat history.

2. Technology Stack

Layer	Technology / Package	Function
Backend Framework	Python 3.13 / Django 6.0.6	Core Web Backend, ORM, Views, Middleware
Database	MySQL / MariaDB (via PyMySQL)	Relational Database Engine (umi_chatbot_db)
Frontend	HTML5, Vanilla JS, Vanilla CSS	Responsive Glassmorphic UI & Dark/Light Themes
RAG Engine	LlamaIndex (llama-index-core)	Document Chunking, Node Parsing & Vector Search
LLM Integrations	google-genai (v2.10.0), Ollama REST API	Gemini 2.5 & Ollama Model Inference Drivers
Image Processing	Pillow (v12.3.0)	Avatar Upload Optimization & Resizing
Email Subsystem	Django SMTP + EmailMultiAlternatives	HTML Password Reset Email Delivery

3. Comprehensive Record of All Changes Made

3.1 MySQL Database Connection & Migration

• PyMySQL compatibility patch injected into settings.py for MariaDB/MySQL support.

• Created clean schema migration script (create_clean_schema.py) initializing 14 clean database tables.

3.2 Enterprise Authentication & Security Subsystem

• Implemented Registration (/register/), Login (/login/), and Logout (/logout/) with Remember Me (30-day session expiry option).

• Created EnterpriseAuthMiddleware protecting private routes and enforcing anti-caching HTTP headers (Cache-Control: no-cache, no-store) to block browser back-button access post-logout.

3.3 Enterprise Password Reset System

• Database Model (PasswordResetToken): Stores cryptographically secure 64-char tokens, 30-min expiry, and is_used status.

• Anti-Enumeration & Rate Limiting: Form accepts Email OR Username and returns anti-enumeration security response. Rate limited to 3 requests per 15 minutes per IP/input.

• SMTP HTML Email Dispatch: Responsive HTML email template dispatched via Gmail SMTP (smtp.gmail.com:587).

• Password Update & Session Invalidation: Updates password using PBKDF2 hashing, marks token as used, and invalidates all active user sessions across browsers.

3.4 Profile Management & Avatar Processing

• Profile Settings: Real-time update of Username, Email, First Name, and Last Name in MySQL.

• Pillow Avatar Processing: Resizes uploaded profile pictures to 300x300 JPG format using LANCZOS thumbnail filtering.

• Dynamic Avatars: Displays user avatar picture or capitalized initials dynamically in header & sidebar.

3.5 Atomic Account Deletion

• Danger Zone Modal: Requires current password verification in a Bootstrap modal.

• Transactional Cascade: Transactionally wipes user attachments from disk, avatar picture, messages, sessions, memory, settings, profile, and user account.

3.6 LlamaIndex RAG & Navigation Fixes

• Integrated LlamaIndex SentenceSplitter and LlamaDocument parser in chat/rag_service.py.

• Restored multi-turn attachment database linking in send_message, resolving PDF explanation errors.

• Expanded file upload extensions to 25+ formats (.pdf, .docx, .txt, .csv, .xlsx, .pptx, .png, .jpg, .code).

• Global New Chat Navigation: Converted sidebar new chat button to explicit URL route, fixing navigation on Settings/Help pages.

4. Database Schema Structure

Table Name	Description & Purpose
users	Core user account table (username, email, PBKDF2 password)
user_profiles	User profile metadata (bio, avatar path, role, account status)
user_settings	User preferences (theme, default AI provider, model, response style)
ai_preferences	Fine-grained AI generation parameters (temperature, max_tokens)
chat_sessions	Conversation metadata (UUID, title, is_pinned, is_archived)
messages	Chat messages (sender, content, timestamp, token count)
conversation_memory	Summarized long-context conversation memory records
documents	Uploaded file attachments (file_name, file_size, path, mime_type)
auth_sessions	Active user login session tracking
analytics	Application metrics (request throughput, latency)
runtime_logs	High-precision system logs and exception tracebacks
password_reset_tokens	Cryptographically secure single-use password reset tokens